



考研竞赛凯哥 第9次周考

考试时间: 5月17日上午 08:30~10:00

讲解时间: 5月17日上午 10:00~12:00

注1: 本次周考的范围为《高数上册》、《多元微分的概念、计算、极值》

注2: 周考, 每周一次, 比较频繁, 所以题量减半, 考试时间减半, 且选填题与解答题的比例随机;

月考, 每月一次, 完全按照考研的标准进行设置.

注3: 无论周考还是月考, 目的都是查漏补缺和巩固所学知识, 所以题目的难度至少持平或略高于真题的平均难度, 大家一定不要因为模考成绩不理想而灰心, 考前犯的任何错误都应该珍惜!

$$1. \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\sqrt{(n+k)(n+k+1)}}{n^2}.$$

解:

$$a_n \leq \sum_{k=1}^n \frac{n+k+1}{n^2} = \sum_{k=1}^n \left(\frac{1}{n} + \frac{k}{n^2} + \frac{1}{n^2} \right) = n \cdot \frac{1}{n} + \frac{\frac{1}{2}n(n+1)}{n^2} + \frac{1}{n} \xrightarrow{n \rightarrow \infty} 1 + \frac{1}{2} + 0 = \frac{3}{2}$$

$$a_n \geq \sum_{k=1}^n \frac{n+k}{n^2} = \sum_{k=1}^n \left(\frac{1}{n} + \frac{k}{n^2} \right) = n \cdot \frac{1}{n} + \frac{\frac{1}{2}n(n+1)}{n^2} \xrightarrow{n \rightarrow \infty} 1 + \frac{1}{2} = \frac{3}{2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = \frac{3}{2}$$

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$$2. \text{ 设 } f(x, y), g(x, y) \text{ 有二阶连续偏导数, } \varphi(x) = f[x, g(x^2, x^2)], \text{ 求 } \frac{d^2 \varphi}{dx^2}.$$

解:

$$\frac{d\varphi}{dx} = f_1' \cdot 1 + f_2' \cdot (g_1' \cdot 2x + g_2' \cdot 2x) = f_1' + 2xf_2'g_1' + 2xf_2'g_2'$$

$$\frac{d^2 \varphi}{dx^2} = \{f_{11}'' \cdot 1 + f_{12}'' \cdot (g_1' \cdot 2x + g_2' \cdot 2x)\}$$

$$+ \left\{ 2f_2'g_1' + 2xg_1' \cdot [f_{21}'' \cdot 1 + f_{22}'' \cdot (g_1' \cdot 2x + g_2' \cdot 2x)] + 2xf_2' \cdot (g_{11}'' \cdot 2x + g_{12}'' \cdot 2x) \right\}$$

$$+ \left\{ 2f_2'g_2' + 2xg_2' \cdot [f_{21}'' \cdot 1 + f_{22}'' \cdot (g_1' \cdot 2x + g_2' \cdot 2x)] + 2xf_2' \cdot (g_{21}'' \cdot 2x + g_{22}'' \cdot 2x) \right\}$$





3. 设 $f(x) = e^{\frac{1}{1+x}}$, 求 $f^{(3)}(0)$.

解:

$$f(x) = e^{\frac{1}{1+x}-1} = e \cdot e^{\frac{-x}{1+x}} \sim (-x).$$

$$\Rightarrow f(x) = e \cdot \left[1 + \frac{-x}{1+x} + \frac{1}{2!} \frac{x^2}{(1+x)^2} + \frac{1}{3!} \frac{-x^3}{(1+x)^3} + o(x^3) \right]$$

$$\text{又由于 } x \rightarrow 0 \text{ 时, } \frac{-x}{1+x} = (-x) \cdot \frac{1}{1+x} = (-x) \cdot [1 - x + x^2 - x^3 + x^4 - \dots]$$

$$= -x + x^2 - x^3 + o(x^3)$$

$$\Rightarrow f(x) = e \cdot \left[1 + (-x + x^2 - x^3) + \frac{1}{2} \cdot (-x + x^2)^2 + \frac{1}{6} \cdot (-x)^3 + o(x^3) \right]$$

$$= e \cdot \left[1 - x + \frac{3}{2}x^2 - \frac{13}{6}x^3 + o(x^3) \right]$$

$$\Rightarrow -\frac{13}{6}e = \frac{f'''(0)}{3!} \Rightarrow f'''(0) = -13e$$

4. 设 $f(x, y)$ 在点 $(1, 1)$ 处连续, 且 $f(x, y) = \ln(1 + x^2 + y^2) + o(\rho)$, $\rho = \sqrt{(x-1)^2 + (y-1)^2}$.

(1) 求 $df(x, y)|_{(1,1)}$; (2) 求 $\lim_{t \rightarrow 0} \frac{f(1+3\sin t, 1) - f(1, 1-3\tan t)}{t}$.

解:

$$(1) f(1, 1) = \lim_{(x,y) \rightarrow (1,1)} f(x, y) = \ln 3 + 0 = \ln 3.$$

$$f'_x(1, 1) = \lim_{x \rightarrow 1} \frac{f(x, 1) - f(1, 1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\ln(2 + x^2) - \ln 3 + o(x-1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\ln \frac{2+x^2}{3}}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{\frac{x^2-1}{3}}{x-1} = \lim_{x \rightarrow 1} \frac{x+1}{3} = \frac{2}{3}.$$

$$\text{同理 (轮换对称)} f'_y(1, 1) = \frac{2}{3} \Rightarrow df(x, y)|_{(1,1)} = \frac{2}{3}dx + \frac{2}{3}dy$$

$$(2) I = \lim_{t \rightarrow 0} \frac{f(1+3\sin t, 1) - f(1, 1)}{3\sin t} - \lim_{t \rightarrow 0} \frac{f(1, 1-3\tan t) - f(1, 1)}{-3\tan t} \cdot (-3)$$

$$= 3 \cdot f'_x(1, 1) + 3 \cdot f'_y(1, 1) = 4$$





5. $\int_0^{+\infty} \frac{1}{1+x^4} dx.$

解:

$$\begin{aligned} I &\stackrel{x=\frac{1}{t}}{=} \int_{+\infty}^0 \frac{1}{1+\frac{1}{t^4}} \cdot \left(-\frac{1}{t^2}\right) dt = \int_0^{+\infty} \frac{t^2}{t^4+1} dt = \int_0^{+\infty} \frac{x^2}{x^4+1} dx \\ &= \frac{1}{2} \int_0^{+\infty} \frac{x^2+1}{x^4+1} dx = \frac{1}{2} \int_0^{+\infty} \frac{\left(1+\frac{1}{x^2}\right)}{x^2+\frac{1}{x^2}} dx = \frac{1}{2} \int_0^{+\infty} \frac{1}{\left(x-\frac{1}{x}\right)^2+(\sqrt{2})^2} d\left(x-\frac{1}{x}\right) \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \arctan \frac{x-\frac{1}{x}}{\sqrt{2}} \Big|_0^{+\infty} = \frac{1}{2\sqrt{2}} \cdot \left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right] \\ &= \frac{\pi}{2\sqrt{2}} \end{aligned}$$

6. 求曲线 $y = x^3 \left(e^{\frac{1}{x}} + e^{-\frac{1}{x}} - 2 \right)$ 的全部渐近线.

解:

竖直: $\lim_{x \rightarrow 0} y = +\infty$, 故 $x=0$ (y 轴) 为一条竖直渐近线.

$$\begin{aligned} \text{斜: } x \rightarrow \infty, y &= x^3 \cdot \left[\left(1 + \frac{1}{x} + \frac{1}{2} \cdot \frac{1}{x^2} + \frac{1}{6} \cdot \frac{1}{x^3} \right) + \left(1 - \frac{1}{x} + \frac{1}{2x^2} - \frac{1}{6x^3} \right) - 2 \right] + o\left(\frac{1}{x^3}\right) \\ &= x^3 \left[1 \cdot \frac{1}{x^2} + o\left(\frac{1}{x^3}\right) \right] = x + o(1) \end{aligned}$$

综上: 斜渐近线为 $y=x$, 竖直渐近线为 $x=0$

7. 设 $f(x,y)$ 满足 $f'_x(x,y) = y(1+x)e^{x-y}$, $f(1,y) = ye^{1-y}$, 求 $f(x,y)$ 的表达式和 $f(x,y)$ 的极值.

解: 在第一个等式两边对 x 偏积分:

$$\begin{aligned} f(x,y) &= \int y(1+x)e^{x-y} dx = y \cdot e^{-y} \int (x+1)e^x dx = ye^{-y}[xe^x + C(y)] \\ \Rightarrow f(1,y) &= ye^{-y} \cdot [1 \cdot e^1 + C(y)] = ye^{1-y} + C(y) \cdot ye^{-y} \Rightarrow C(y) = 0 \\ \Rightarrow f(x,y) &= ye^{-y} \cdot xe^x \end{aligned}$$

$$\text{令 } \begin{cases} f'_x = 0 \\ f'_y = 0 \end{cases} \Rightarrow \begin{cases} ye^{-y} \cdot (x+1)e^x = 0 \\ xe^x \cdot (1-y)e^{-y} = 0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=0 \end{cases} \text{ 或 } \begin{cases} x=-1 \\ y=1 \end{cases} \Rightarrow P_1(0,0), P_2(-1,1)$$

(1) $P_1(0,0)$, $f(0,0)=0$, 但 $f(x,y)$ 在 $(0,0)$ 去心邻域有正有负 $\Rightarrow f(0,0)$ 不是极值

$$\begin{aligned} (2) \text{ 在 } P_2(-1,1) \text{ 处: } &\begin{cases} f_{xx}'' = ye^{-y} \cdot (x+2)e^x \\ f_{yy}'' = xe^x \cdot (y-2)e^{-y} \\ f_{yx}'' = (x+1)e^x \cdot (1-y)e^{-y} \end{cases} \xrightarrow{x=-1, y=1} \begin{cases} A = 1 \cdot e^{-1} \cdot 1 \cdot e^{-1} = e^{-2} \\ C = -1 \cdot e^{-1} \cdot (-1)e^{-1} = e^{-2} \\ B = 0 \end{cases} \\ \Rightarrow AC - B^2 &= e^{-2} \cdot e^{-2} - 0^2 = e^{-4} > 0 \text{ 且 } A > 0 \Rightarrow \text{极小值为 } f(-1,1) = -e^{-2} \end{aligned}$$





8. 设 $f(x)$ 在 $[0, 1]$ 连续, $f(0)=0, f(1)=1$, 证: 对 $\forall n \geq 2$, 存在 $\xi \in [0, 1]$, 使得 $f\left(\xi - \frac{1}{n}\right) = f(\xi) - \frac{1}{n}$.

解:

$$\text{令 } F(x) = f\left(x - \frac{1}{n}\right) - f(x) + \frac{1}{n}$$

$$\begin{cases} F\left(\frac{1}{n}\right) = f(0) - f\left(\frac{1}{n}\right) + \frac{1}{n} \\ F\left(\frac{2}{n}\right) = f\left(\frac{1}{n}\right) - f\left(\frac{2}{n}\right) + \frac{1}{n} \\ F\left(\frac{3}{n}\right) = f\left(\frac{2}{n}\right) - f\left(\frac{3}{n}\right) + \frac{1}{n} \\ \vdots \\ F\left(\frac{n}{n}\right) = f\left(\frac{n-1}{n}\right) - f(1) + \frac{1}{n} \end{cases}$$

$$\text{相加: } F\left(\frac{1}{n}\right) + F\left(\frac{2}{n}\right) + \cdots + F\left(\frac{n}{n}\right) = f(0) - f(1) + n \cdot \frac{1}{n} = 0 - 1 + 1 = 0$$

(1) 若 $F\left(\frac{1}{n}\right) = F\left(\frac{2}{n}\right) = \cdots = F\left(\frac{n}{n}\right) = 0$ 此时, 取 $\xi = \frac{1}{n}, \frac{2}{n}, \dots, \frac{n}{n}$ 均可!

(2) 若 $F\left(\frac{k}{n}\right)$ ($k=1, 2, \dots, n$) 有正有负, 则由零点定理, $\exists \xi, s.t. F(\xi) = 0$.

9. 求 $|z|$ 在约束条件 $\begin{cases} x^2 + 9y^2 - 2z^2 = 0 \\ x + 3y + 3z = 5 \end{cases}$ 下的最大值与最小值.

解:

$|z|$ 与 z^2 有相同的极值点.

$$\text{令 } L(x, y, z, \lambda, \mu) = z^2 + \lambda(x^2 + 9y^2 - 2z^2) + \mu(x + 3y + 3z - 5).$$

$$\text{令 } \begin{cases} L_x' = 0 \\ L_y' = 0 \\ L_z' = 0 \\ L_\lambda' = 0 \\ L_\mu' = 0 \end{cases} \Rightarrow \begin{cases} 2\lambda x + \mu = 0 \\ 18\lambda y + 3\mu = 0 \\ 2z - 4\lambda z + 3\mu = 0 \\ x^2 + 9y^2 - 2z^2 = 0 \\ x + 3y + 3z = 5 \end{cases} \Rightarrow x = 3y \Rightarrow \begin{cases} x = 1 \\ y = \frac{1}{3} \\ z = 1 \end{cases} \text{ 或 } \begin{cases} x = -5 \\ y = -\frac{5}{3} \\ z = 5 \end{cases}$$

$$\Rightarrow |z|_{\min} = 1, |z|_{\max} = 5$$

10. 设 $f(x)$ 在 $[0, 2]$ 上二阶可导, $|f(x)| \leq 1, |f''(x)| \leq 1 (0 \leq x \leq 2)$, 证明: $|f'(x)| \leq 2$.

解:

$$\begin{cases} f(0) = f(x) + f'(x) \cdot (0 - x) + \frac{f''(\xi_1)}{2!} (0 - x)^2 \\ f(2) = f(x) + f'(x) \cdot (2 - x) + \frac{f''(\xi_2)}{2!} (2 - x)^2 \end{cases}, \quad 0 < \xi_1 < x < \xi_2 < 2$$

$$\frac{f(2) - f(0)}{2} = f'(x) + \frac{1}{4} [f''(\xi_2)(2 - x)^2 - f''(\xi_1)x^2]$$





$$\Rightarrow |f'(x)| \leq \frac{1}{2}(1+1) + \frac{1}{4} \cdot [1 \cdot (2-x)^2 + 1 \cdot x^2]$$

$$= 1 + \frac{1}{4}(x^2 + (2-x)^2) \leq 1 + \frac{1}{4} \cdot 4 = 1 + 1 = 2$$

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